V-M-96G
April 11, 1975
SUPERSEDING
Int. Fed. Spec. V-M-0096F
June 19, 1974
(see 6.6)

FEDERAL SPECIFICATION

MATTRESSES, BED, INNERSPRING

This specification was approved by the commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the construction and materials for innerspring bed mattresses.

1.2 Classification.

1.2.1 Types. The mattresses shall be of the following types, classes, sizes and grades as specified (see 6.2):

Type I - Springs in individual pockets.
Type II - All wire spring units, free end coil construction.
Type III - All wire spring units, knotted coil construction.
Type IV - All wire spring units, offset construction.

1.2.2 Classes.

Class 1 - Without handles.
Class 2 - With handles.

1.2.3 Sizes. Sizes of mattresses shall be as shown in tables II, III and IV, as specified (see 6.2).

1.2.4 Grades.

Grade A - 50 percent cotton, 50 percent linters.
Grade B - 65 percent polyester, 35 percent linters.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 7210

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Federal Specifications:

V-T-276	-	Thread, Cotton
V-T-285	-	Thread, Polyester
CCC-C-432	-	Cloth, Sheeting, Cotton, (Unbleached, Bleached, and Dyed)
CCC-C-436	-	Cloth, Ticking Twill, Cotton
DDD-L-20	-	Label; for Clothing, Equipage, and Tentage, (General Use)
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies)
Fed. Std. No. 191 - Textile Test Methods
Fed. Std. No. 751 - Stitches, Seams, and Stitchings

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outline under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-41826	-	Batting, Synthetic Fibers: Polyester, (Quilted and Unquilted)
MIL-M-3996	-	Mattresses (Except Pneumatic Type), Packaging and Packing Of

Military Standard:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Consumer Product Safety Commission

FF-4-72 - Flammability Standard for Mattresses

(Application for copies should be addressed to Consumer Product Safety Commission, 7315 Wisconsin Avenue (East Tower), Bethesda, MD 20207.)

US Department of Agriculture Publication

C&MS-SRA-180 - Cotton and Cotton Linters Standards

(Application for copies should be addressed to the Department of Agriculture, Washington, DC 20250).

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

3.2 Flammability. All mattresses shall meet the provisions of the Mattress Flammability Standard FF-4-72 (see 4.6).

3.3 Materials.

3.3.1 Cotton. The cotton shall be new (not previously used) minimum, 7/8-inch staple cotton and not lower in grade than the lowest grade in the white, spotted, or gray grades, conforming to the standard of the United States Department of Agriculture Publication No. C&MS-SRA-180 (see 2.2). The use of reginned cotton will be permitted provided the required cotton is equivalent in grade to the standards listed above. Tinged or stained cotton shall not be permitted. Testing shall be as specified in 4.4.1.

3.3.2 Linters. The linters for the mattresses shall be new (not previously used) conforming to grade number 3 and staple 3 of USDA Publication No. C&MS-SRA-180 (see 2.2). Testing shall be as specified in 4.4.1.1.

3.3.3 Polyester fiber. The fiber to be blended with the linters specified in 3.3.2 for grade B felt (see 3.3.4.2) shall consist of a minimum range of 20 to 30 percent mixture of crimped, (6 to 8 crimps per inch) cut staple virgin polyester fiber, of 5.0 or 6.0 denier, 2 ± 1/2 inch in length and the balance of the blend may be made from polyester fiber (95 percent pure polyester fiber) (see 4.4.1.2).

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3.3.4 Felt. The felt shall be clean and shall be so processed as to produce unbroken laminated sheets.

3.3.4.1 Grade A. Grade A felt shall be made of a blend of 50 percent cotton conforming to 3.3.1 and 50 percent linters conforming to 3.3.2.

3.3.4.2 Grade B. When specified (see 6.2) grade B felt shall be made of a blend of 65 percent polyester fiber conforming to 3.3.3 and 35 percent linters conforming to 3.3.2 (see 4.4.1.2).

3.3.5 Barrier materials.

3.3.5.1 Polyester batting. Polyester batting conforming to type III batting of MIL-B-41826 shall be used as a barrier, except that the requirements for thickness after laundering, compressional recovery after laundering, and launderability need not apply.

3.3.5.2 Commercial bonded polyester batting. The use of a commercial bonded polyester batting for use as a barrier material is acceptable provided that the batting is bonded with an approved durable bonding agent (see 6.5) uniformly applied. The maximum concentration of the applied bonding agent shall not exceed 18.0 percent on the weight of the finished bonded batting. The bonding agent shall be applied to both sides of the batting filler.

3.3.5.3 Approval and qualifications of other barrier materials. Other commercial type mattress barrier materials or smoldering resistant treatments can also be used as barrier materials. Approval of these materials and or treatments must be obtained prior to use (see 6.4). The use of an impermeable material as a barrier material is prohibited (see 4.4.1).

3.3.6 Ticking. The ticking shall conform to the requirements of type I or II, class 1 or 2, ticking of CCC-C-436.

3.3.7 Sheeting. The cotton sheeting shall conform to type VII or VIII, class 1 of CCC-C-432.

3.3.8 Open mesh or non-woven fabric. The open mesh fabric shall have a weight of 1-1/2 ounces to 3-1/2 ounces per square yard and shall be made from a thermoplastic material. The non-woven fabric shall have a weight of 1-1/2 ounces to 3-1/2 ounces per square yard and shall be either a spun bonded nylon, polyester or polypropylene fabric (see 3.5.11.1 and 4.4.1.2).

3.3.9 Heat perimeter shield or flange. The material used for flanging or a heat perimeter shield shall be an aluminized fabric (see 4.4.1.2).

3.3.10 Thread. The thread used for stitching the edge binding tape and the border shall be cotton, ticket number 30, 3 ply, type IA1 or IA2 of V-T-276 or the thread shall be polyester; type I, class 1, sub-class A, size E of V-T-285. The thread used for quilting shall be polyester conforming to type I or II, size B, class 1, sub-class A of V-T-285.

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3.3.11 Tape.

3.3.11.1 Edge binding tape. The tape used in finishing the edges of the mattress shall be 5/8 inch (minimum) width polyester tape or an approved fire resistant or smoldering ignition resistant treated cotton tape (see 6.4). The polyester tape shall have a minimum breaking strength of 100 pounds and the treated cotton tape shall have a minimum breaking strength of 30 pounds. The color of the tape shall be white or natural. Testing shall be as specified in 4.4.1.1.

3.3.11.2 Tufting tape, braid or twine. The tape braid or twine used for tufting shall be either nylon or polyester. The nylon or polyester tape shall be not more than 5/8 inch wide and shall have a 50 pound minimum breaking strength. The breaking strength for the nylon or polyester tufting braid or twine shall be 80 pounds minimum. In addition to nylon or polyester tape innertufted mattresses can use cotton tape, braid or twine for tufting. The cotton tape shall be not less than 3/8 inch nor more than 5/8 inch wide and shall have a minimum breaking strength of 50 pounds. The cotton tufting braid or twine shall have a breaking strength of not less than 32 pounds. Testing shall be as specified in 4.4.1.1.

3.3.12 Handles.

3.3.12.1 Class 1. Class 1 mattresses shall be furnished without handles.

3.3.12.2 Class 2. When specified (see 6.2), class 2 mattresses shall be furnished with two horizontal handles (lifting straps) attached to the border on each side of the mattress. The handles shall be a commercial, braid covered, cotton or synthetic cord lifting strap utilizing nonferrous eyelets or grommets, metal clip cord ends and a spring steel reinforcing strip for each handle. The diameter of the braid covered cord and the finished length of the handles and the spacing of the grommets or eyelets shall be consistent with good commercial practices (see 4.4.1.2).

3.3.13 Unicellular polyethylene foam. The border of the mattress can be constructed by using unicellular polyethylene foam. The foam shall be 1/4 inch thick, 5 $\pm$ 1/4 inches wide conforming to the requirements of PPP-C-1752, except that the requirements for compression set, contact corrosivity, dusting and breakdown, abrasiveness, thermal stability and stress - strain stability need not apply.

3.3.14 Tufting buttons. The tufting buttons shall be a commercial type, of the size, strength, shape and finish commonly used in mattress construction. The color of the tufting buttons shall be either white, blue or egg-shell (grayish white). All tufting buttons contained in a mattress shall be of the same color, i.e., all white, or all blue, or all egg-shell (see 4.4.1.2).

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3.3.15 Ventilators. The ventilators shall be of one or two piece construction, of brass or aluminum not less than 0.013 inch thick, with an outside diameter of 7/8 to 1 inch. The length of the ventilator shank shall be of sufficient length to effect a secure clinch of the border without cutting the surrounding ticking. The diameter of the shank shall be 11/16 to 3/4 inch. A minimum of 61 percent of the face of the ventilator shall be perforated with not less than 61 holes 1/32 to 1/16 inch in diameter (see 4.4.1.2).

3.3.16 Wire mat. When wire mat is used as the spring covering specified in 3.5.5.2, it shall conform to the following: The wire mat shall have a series of steel wires spaced 1 inch ± 3/8 inch apart, running at right angles to longitudinal fiber cords which are uniformly spaced not more than 8 inches apart. The steel wires shall pierce and pass through the center cords and shall be crimped around the border cords. The wire shall be 19 gage oil-tempered steel wire with a tensile strength range of 266,000 p.s.i. to 296,000 p.s.i. The spirally wound fiber cords shall be constructed of not less than 30 pound kraft paper. The center cords shall be not less than 0.16 inch in diameter, with or without a core, and shall have a breaking strength of not less than 100 pounds. The border cords shall be not less than 0.125 inch in diameter with a breaking strength of not less than 100 pounds. The dimensions of the wire mat shall be 3 inches ± 1/4 inch less in length and 2 inches ± 1/4 inch less in width of the dimensions of the applicable spring unit used for the end item mattress. Testing of the components of the wire mat shall be as specified in 4.4.1.2. Inspection of the completely fabricated wire mat shall be as specified in 4.4.2.4.

3.3.17 Hog rings. The hog rings (see 3.5.5.2) shall be commercial type, of the size, strength and finish commonly used in mattress construction (see 4.4.1.2).

3.3.18 Coils.

3.3.18.1 Coils for type I mattress. The wire for the spring coils shall be not less than 16 gage wire. The wire shall be hard-drawn, spring-steel wire with a minimum tensile strength of 240,000 p.s.i. and a maximum tensile strength of 280,000 p.s.i. The wire shall have a protective zinc coating which will not be visibly affected by salt spray, nor will there be visible evidence of rust when tested as specified in 4.4.1.1. The coils shall be 2-1/2 ± 1/16 inch in diameter, cylindrical or barrel shaped, and shall have both ends of wire turned in. There shall be not less than 3 nor more than 3-1/2 turns per coil, measured on the free coil, not including end bearing convolutions (see 4.2.1.2).

3.3.18.2 Coils for type II mattress. The wire for the edge and corner coils shall be not less than 13 gage wire. The wire shall be hard-drawn, spring-steel wire with a minimum tensile strength of 205,000 p.s.i. and a maximum tensile strength of 250,000 p.s.i. The wire for all other coils shall be not less than 13-1/2 gage wire. The wire shall be hard-drawn, spring-steel wire with a minimum tensile strength of 212,500 p.s.i., and a maximum tensile strength of 275,500 p.s.i., when tested as specified in 4.4.1. Corner coils shall be cylindrical,

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with not less than 6 convolutions. The ends of the corner coils shall be die formed to 4 inch ± 1/2 inch radius. End coils shall be offset cylindrical or barrel overlapping-type coils. Edge coils forming the sides of the spring unit shall present a straight edge to the outer edge of the spring unit and shall be cylindrical. All other inside coils shall be hour-glass shape. All coils except the corner coils shall have not less than 4-1/2 active turns or complete convolutions. All coils shall be free-end type coils to allow their being anchored within the helical without accessory locking device. The coils shall be 5 inches ± 1/4 inch in height. Coils or the complete spring unit assembly shall be given suitable thermal threatment to relieve residual stresses caused by coiling (see 4.4.1.2). Coils shall not decrease in height more than 3/16 inch when tested as specified in 4.4.1.1.

3.3.18.3 Coils for type III mattress. The wire for the spring coils shall be not less than 13 gage wire. The wire shall be hard-drawn, spring wire with a minimum tensile strength of 205,000 p.s.i., and maximum tensile strength of 250,000 p.s.i. All coils shall be hourglass in shape, except that the coils in the two longitudinal rows forming the sides of the spring unit may be cylindrical or elliptical in shape. There shall be not less than three active turns or convolutions per coil, including only portions between the knots of the coil, and the coils shall be 5 inches ± 1/4 inch in height. Coils or the complete spring unit assembly shall be given suitable thermal treatment to relieve stresses caused by coiling (see 4.4.1.2). Coils shall not decrease in height more than 3/16 inch when tested as specified in 4.4.1.1.

3.3.18.4 Coils for type IV mattress. The wire spring for the coils for the type IV mattresses shall be as specified for the wire spring coils of the type III mattresses (see 3.3.18.3). All coils shall be hour-glass shaped (except edge coils which shall be cylindrical) and shall have two precision offset sections in each convolution of the coil, die formed to accurately place the coil in its proper transverse position to receive the interlocking helical which surrounds each offset portion of the coil and securely holding each coil in position to permit complete freedom of hinging action. There shall be not less than three active turns or convolutions per coil. The coils shall be 5 inches ± 1/4 inch in height. Coils or the complete spring unit assembly shall be given suitable thermal treatment to relieve residual stresses caused by coiling (see 4.4.1.2). Coils shall not decrease in height more than 3/16 inch when tested as specified in 4.4.1.1.

3.3.19 Innerspring units.

3.3.19.1 Type I mattress (springs in individual pockets). The coils specified in 3.3.18.1 shall be contained in a continuous string of fabric pockets made from sheeting specified in 3.3.7 or non-woven fabric specified in 3.3.8 each pocket containing one coil. The coils shall stand upright and shall be parallel to one another. The coils shall be not less than 4 inches in height when confined only by the pockets. The side stitching between individual pockets for each coil shall be done with one row of machine stitching extending from top to bottom of pocket. The top of pockets shall be seamed closed by machine. The pockets

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shall be such as to accomodate the coil, leaving only sufficient slack to permit proper spring action. Each coil shall be held in the pocket at all times in a manner to insure positive pressure against both the top and bottom layers of felt in the mattress, so as to provide against the development of slackness or looseness between the spring and the top and bottom felt layers. The innerspring assembly shall be built by alternately weaving the continuous rows of pocketed springs lengthwise. Each row of coils shall nest between the adjacent row in a manner to allow overlapping of the coils, and provide against any open spaces between the pocketed coils. After being inclosed in pockets and laid up in mattress formation, the individual spring coils shall be tied together at the ends or through the center of the pocketed coils in the direction of the width of the assembly with rows of twine spaced not more than 9 inches apart in the direction of the length.

3.3.19.2 Type II mattress (all-wire spring unit, free-end coil construction). The spring unit shall be made with materials specified in 3.3.18.2, 3.3.20 and 3.3.21 located so as to keep edges from sagging, prevent side sway, to give firm support to all parts of the body, and to maintain flexibility. All-wire spring units, free-end coil construction, shall consist of transverse rows of alternating right-hand and left-hand coils connected with a long helical specified in 3.3.21 running at right angles to the long axis of the unit in such a manner as to prevent lateral slippage. The free end of the coil shall be anchored within the helical without the use of any accessory locking device. The edges of all outside coils shall be constructed so as to present a straight edge to the construction. The individual coils forming the end rows shall have a double overlap of the end convolutions into the end helical and second helical. The end rows shall have a stabilizer wire specified in 3.3.20 placed within each of the 4 end helicals with the free ends of wire extending to a point on line with the center of the second row of coils at each end, and fastened so as to prevent slippage. The 4 free-end corner coils of the unit construction shall be cylindrical. The ends of the helicals shall be turned in to form a smooth, rounded termination. The spring unit shall be furnished with a baked-on enamel, or rust-resisting metallic or aluminum coating.

3.3.19.3 Type III mattress (all wire spring unit, knotted coil construction). All-wire spring units, knotted-coil construction, shall consist of transverse rows of coils specified in 3.3.18.3, 3.3.20 and 3.3.21 connected by helicals running at right angles to the long axis of the unit to prevent lateral slippage and to permit a free-hinge perion when the mattress is folded or rolled from end to end, or shall consist of transverse rows of individual coils or units of not more than 4 coils to a unit, joined to the neighboring coils or units of coils to prevent lateral slippage and to permit a free-hinge action when the mattress is folded from end to end when tested as specified in 4.4.6. Where helicals are used in the construction, the individual spring coils may be offset with a portion or portions passing through not less than three turns of the helical. All knots shall be turned down and the ends of all helicals shall be turned inward forming a smooth and rounded termination. The unit shall be finished with a baked-on enamel, or rust-resisting metallic or aluminum coating.

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3.3.19.4 Type IV mattress (all wire spring unit offset construction). The spring unit shall be made with materials specified in 3.3.18.4, 3.3.20 and 3.3.21 and shall consist of transverse rows of coils connected by helicals running at right angles to the long axis of the unit. The die formed offset sections shall receive the interlocking helicals so as to securely hold each coil in proper position and prevent lateral slippage of the coils in the assembled unit so as to maintain dimensional stability, balance of the spring action of each coil operates with a visible degree of independence. The spring unit shall be finished with a baked-on enamel, or rust-resisting metallic or aluminum coating.

3.3.20 Stabilizer wire (end rods) (types II, III, and IV). The stabilizer wire shall be made from high-carbon steel, oil-tempered, 15 gage to 10-1/2 gage wire, when tested as specified in 4.4.1. The corners shall be die-formed to a 4 inch ± 1/2 inch radius. The stabilizers shall be attached to the outside of each end row of coil so as to prevent slippage, and shall be formed to fit the full width of the spring unit and extend around the corners of the unit at least ten inches (see 4.4.1.2).

3.3.21 Helicals for type II (free end coil construction), type III (knotted coil construction) and type IV (offset construction) mattresses. The helical wire shall be made from hard-drawn high-carbon steel wire, not less than 18 gage, when tested as specified in 4.4.1. The formed helical shall be not more than .350 inch outside diameter and shall have a 15/32 inch pitch or less. The helical shall have not less than twenty-four turns per foot. All helicals shall be given suitable thermal treatment to relieve residual stresses caused by coiling (see 4.4.1.2).

3.4 Design. Mattresses shall be either outertufted or innertufted or quilted. The detailed construction requirements shall be as specified in 3.5.

3.5 Construction. The construction requirements for all types of mattresses shall be as specified herein.

3.5.1 Stitching. Stitching shall conform to stitch type 301 or 401 of FED-STD-751 with 8 to 12 stitches per inch (except for quilting). When 401 stitching is used, the chain portion of the stitching shall not appear on the outside of the ticking.

3.5.2 Seams. When longitudinal pieces of ticking are used they shall be stitched with stitch type 401 and seamed with seam type LSb-2 or stitch type 301 and seam type SSa-1 conforming to FED-STD-751.

3.5.3 Type 301 stitching. Ends of all seams and stitching when not caught in other seams and stitching shall be backstitched or overstitched 3/4 inch minimum, except for closing the mattress on a tape edge machine. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

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3.5.3.1 Repairs or type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 3/4 inch back of the end of the stitching. 1/

b. Thread breaks, two or more consecutive skipped stitches or run-offs noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 3/4 inch in back of the defective area, continue over the defective area and continue a minimum of 3/4 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.5.4 Types 401, 504 and 505. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.3.1(a) and 3.5.3.1(b) and type 301 stitching may be used for making the repairs of type 401 stitching.

3.5.5 Insulation for innerspring unit.

3.5.5.1 Type I. Type I mattresses shall have a fitted insulation cover over the innerspring unit. The upper and lower surfaces of the innerspring assembly shall be protected by covers made from sheeting specified in 3.3.7 or by the non-woven fabric specified in 3.3.8. Covers are to overlap all edges of innerspring construction and shall be attached to the spring construction and assembled border all around by a commercial stitching twine. The twine shall be securely knotted at the first side coil and sewn continuously in each side coil and every other end coil around the unit with the twine being securely knotted on the last end coil adjacent to the beginning point.

3.5.5.2 Type II, III or IV. The innerspring units of type II, III or IV mattresses shall be covered by either sheeting specified in 3.3.7 or by the non-woven fabric specified in 3.3.8 or by the wire mat specified in 3.3.16. The sheeting or non-woven fabric shall be attached to both the top and bottom surfaces of the innerspring unit in accordance with the supplier's commercial practice. The wire mat shall be securely fastened to the innerspring unit by three commercial hog rings attached along each longitudinal side. Two of the hog rings shall be located near the ends of the spring unit. Twelve hog rings (total) are required to fasten the wire mat to the spring unit. The hog rings shall not be clamped around the fiber cord on the edge of the wire mat, but shall be clamped around one wire of the mat and the top of the border coil.

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3.5.6 Felt padding for spring units. The springs and insulation shall be padded with felt specified in 3.3.4.1 or when specified (see 6.2) the springs and insulation shall be padded with felt specified in 3.3.4.2. The felt shall be placed over the spring unit and cover assembly in full length and full width layers on both surfaces of the mattress. The felt shall be distributed in layers of equal weight and uniform thickness to adequately cover all areas including edges, ends, and corners. A special crown of felt consisting of 5 to 15 percent of the total weight of felt on each face shall be uniformly distributed over the middle one-third of the length face and for the full width of the mattress. The corners shall be well padded and firm. Both sides of the mattress shall be finished so that either side may be used as the sleeping side with equal comfort.

3.5.7 Barrier material (toppers). The supplier shall use a barrier system or an approved flame retardant or smoldering ignition resistant treatment to meet the mattress flammability provisions specified in 3.2. When a barrier system is used, the barrier materials specified in 3.3.5 shall be placed directly over the felt padding. Barrier materials can also be quilted to the ticking fabric for quilted top mattresses (see 3.5.12). Innertufted mattresses can use a tufting support fabric (see 3.3.8) over the barrier materials. The use of a tufting support fabric (see 3.3.8) for outer tufted mattresses is optional. When approved treatments are to be applied to the felt padding the use of a barrier system or tufting support fabric is optional.

3.5.8 Cover construction. The top and bottom sections in the mattress cover shall each be made up of not more than two longitudinal pieces of ticking specified in 3.3.6. The ticking shall be matched and sewn together (see 3.5.2). The blue stripes of the ticking shall run the lengthwise direction of the mattress. The top and bottom panels shall have hemmed edges when flanging or a heat perimeter shield is not used.

3.5.8.1 Flange or heat perimeter shield attachment. When a flange or heat perimeter shield is used, it shall be 1-3/8 inches to 5-1/2 inches wide and shall be the aluminized fabric specified in 3.3.9. The flange or heat perimeter shield shall be attached to the ticking by using one of the following operations:

(a) The flange or heat perimeter shield shall be attached by overedge stitching it to the raw edge of the ticking.

or

(b) By superimposing the flange or heat perimeter shield on the ticking and seaming with a row of stitches 1/16 to 1/8 inches from the edge.

or

(c) By prehemming the ticking and seaming the flange or heat perimeter shield on the ticking with a row of stitches 1/16 to 1/8 inches from the edge or the operation may be accomplished simultaneously.

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The flange or heat perimeter shield shall also be stitched to the ticking with a second row of seaming stitches (stitch type 301 or 401). This row of stitching shall be positioned along the opposite long edge of the flange or heat perimeter shield material and shall be sewn a minimum of 1/4 inch from the edge. Both rows of stitching of the heat perimeter shield or flange, after overstitching or backstitching the ends of stitching, may be carried off the side edge of the ticking cover and then trimmed. When 401 stitching is used, the chain portion of the stitching shall not appear on the outside of the ticking. When overedge stitching is used it shall be accomplished with stitch type 504 or 505, 3/16 inch gauge of throw, 5 to 7 stitches per inch. The second row of seaming stitches shall have 5 to 7 stitches per inch. Thread shall be as specified in 3.3.10.

3.5.9 Border construction. The border shall be prebuilt prior to attachment to cover. The padding for the border shall be either the felt specified in 3.3.4 or polyester batting specified in 3.3.5 or the unicellular polyethylene foam specified in 3.3.13. The use of other approved barrier materials (see 3.3.5.2) as padding in the construction of the border is also allowed. The border shall be constructed by using the ticking specified in 3.3.6 along with one of the padding materials specified above. The border may be constructed with or without the use of a backing material, as indicated below. The outside of the border shall be made of ticking with both edges hemmed with machine stitching type EFa-1, with a minimum 1/4 inch hem; or the ticking, padding and with or without a backing material may be sewn together to join the border in one operation. The stitching shall go along perpendicular to one folded edge of ticking and run parallel across to form a hem approximately 1-5/8 inches and then run along perpendicular to the opposite folded edge of ticking going parallel again to form a hem approximately 1-5/8 inches. This operation will result in an alternating stitch and skipped hemmed edge along both sides for the entire length of the mattress border. The border on each mattress shall be made of not more than two pieces of ticking with the ends spliced and sewn together with seam type SSa-1, in a continuous manner so as to have no break in its uniformity and finish. The ticking shall be padded with padding material and the width of padding shall be the width of the hemmed border plus 1/4 inch or minus 1/2 inch. The padding shall be butted or spliced at the seam joining the ends of the border in such a manner as to leave no breaks in its uniformity and finish. The felt paddings specified in 3.3.4.1 and 3.3.4.2 are required to have a backing material. The use of backing materials with borders constructed with polyester batting or unicellular polyethylene foam is optional. When a backing material is used on the border, the material shall either be sheeting specified in 3.3.7 or ticking specified in 3.3.6 or the nonwoven fabric specified in 3.3.8 or the aluminized flange and heat shield fabric specified in 3.3.9. The width of the backing material shall be at the option of the supplier. The border shall be sewn together with not less than six stitches per inch. The border shall be equipped with ventilators specified in 3.3.15 and in tables II, III and IV. The ventilators shall be placed in a line approximately half way between the top and bottom hem of the border and shall be secured to border at reasonably spaced intervals (approximately equidistant) on the longitudinal sides of the border. Ventilators shall pass completely through the entire

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border from the outside. When class 2 mattresses are specified the border shall be equipped with two horizontal handles specified in 3.3.12 on each side of the mattress border and located approximately one third of the length from each end of the mattress. The felt padding specified in 3.3.4.1 shall weigh not less than 3.0 ounces per linear yard (see 4.4.1.2). The polyester batt padding specified in 3.3.5 shall weigh not less than 10 ounces per square yard when used in borders (see 4.4.1.2). The entire border shall be constructed so as not to interfere with the free passage of air through the ventilators.

3.5.10 Edge construction. The finished edge of the cover ticking and the finished edge of the border shall be joined together and sewn within the folds of the edge binding tape specified in 3.3.11.1. The tape shall extend completely around both edges of the mattress. Not more than three pieces of tape shall be used on each border edge.

3.5.11 Tufting. Mattresses shall be designed to be either outertufted or innertufted. Tufting shall be accomplished using either button tufts specified in 3.3.14 or tufting twine, braid or tape specified in 3.3.11.2. The depth of tufts shall be 4-1/2 inches to 5-1/2 inches in length. The number of tufts and the pattern shall be as specified in table I. Mattresses shall be tufted through both sides. Knots of tufts shall be arranged so that they will not appear above the surface of either side of the mattress.

3.5.11.1 Innertufted mattresses. Mattresses can also be innertufted resulting in a smooth surface mattress with concealed innertufting. The tufting requirement specified in 3.5.11 shall also apply to innertufted mattresses. Innertufted mattress may require the use of a tufting support fabric to properly anchor the filling materials. The tufting support fabric shall be as specified in 3.3.8. Any tear, rip or cut in the tufting support fabric shall be repaired prior to filling the mattress in the ticking cover.

3.5.12 Quilitng. Matresses can be designed to have a quilted top and bottom surface. The mattress shall be quilted using a commercial quilt pattern. The quilted surface mattress shall be furnished without fabric splices (joining of ticking outer covering) and without areas evidencing complete bobbin changes and shall be consistent with good commercial quality. The quilting shall have 5 to 9 stitches per inch. The filling materials of type II quilted surface mattress shall be anchored by inner tufting. The innertufted requirements specified in 3.5.11. shall also be applicable for type II quilted surface mattresses. The filling materials of types I, III and IV quilted surface mattresses shall be anchored in accordance with good commercial practices.

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TABLE I. Tufting requirements

Size	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Number of tufts 1/	17	28	28	28	28	28	28	28	39	39	39	39	28	28	39	39
Pattern 2/	2x6	3x6	3x6	3x6	3x6	3x6	3x6	3x6	4x6	4x6	4x6	4x6	3x6	3x6	4x6	4x6
Innertufted mattresses 3/																

1/ The spacing of the tufts shall be uniform. For mattresses of 80 inches or longer, there shall be an additional tuft in each lengthwise row for every 10 inch increase in length.

2/ The arrangement of the tufts shall be in a diamond pattern. The distance of any row closest to an edge shall be 6 inches plus or minus 1 inch and the distance of any row closest to an end shall be 6 inches minus 1 inch, plus 2 inches.

3/ The number of tufts, pattern and spacing requirements are applicable to innertufted mattresses.

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3.5.13 Corners. The four corners of the mattresses shall be firm and well padded. The four corners of each mattress shall be rounded on a 4-1/2 ± 1/2 inch radius.

3.6 Physical requirements. Mattresses shall conform to the requirements specified in tables II, III, and IV.

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TABLE II. Physical requirements - Type I

Size	Length (inches) ± 1/2 inch	Width (inches) ± 1/2 inch	Thickness (inches) ± 1/2 inch	Weight pounds min 2/	Number of ventilators Each side	No. of coils in row: Length (min)	Width (min)	Weight of felt on each face pounds 1/
1	76	29	6-1/2	37	4	31	15	7-3/4
2	78	33	6-1/2	44	4	32	17	8-3/4
3	76	34	6-1/2	48	4	31	18	8-3/4
4	75	36	6-1/2	50	4	31	18	9-1/4
5	78	36	6-1/2	52	4	32	18	9-1/2
6	84	36	6-1/2	51	4	33	18	10-1/4
7	75	38	6-1/2	54	4	31	19	9-1/2
8	78	39	6-1/2	51	4	32	19	10-1/4
9	78	41-1/2	6-1/2	56	4	32	20	10-3/4
10	78	47	6-1/2	64	4	32	22	12
11	73	53	6-1/2	64	4	30	24	13-1/4
12	75	53	6-1/2	72	4	31	24	13-1/4

1/ Not a specification requirement included for information purposes.

2/ Not applicable to mattresses using grade B felt (see 4.4.4.1.3 and 6.2).

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TABLE III. Physical requirements - Type II

Size	Length inches ± 1/2 inch	Width inches ± 1/2 inch	Thickness inches ± 1/2 inch	Weight pounds (min)	No. of ventilators Each side	Total No. of coils (min)	Weight of felt layer on each face pounds 1/
1	76	29	6-1/2	30	4	126	8-1/2
2	78	33	6-1/2	39	4	152	10
3 2/		-	-	-	-	-	-
4	75	36	6-1/2	40	4	162	10-1/2
5	78	36	6-1/2	41	4	171	11
6	84	36	6-1/2	47	4	180	12
7	75	38	6-1/2	45	4	162	11-1/2
8	78	39	6-1/2	47	4	171	12
9	78	41-1/2	6-1/2	50	4	190	13
10	78	47	6-1/2	57	4	209	15
11	73	53	6-1/2	58	4	221	16
12	75	53	6-1/2	60	4	234	16
13	80	36	6-1/2	44	4	171	11
14	80	38	6-1/2	48	4	171	12-1/2
15	80	53	6-1/2	65	4	247	17-1/2
16	80	60	6-1/2	65	4	247	17-1/2

1/ Not a specification requirement, included for information purposes.

2/ Size 3 is not applicable to type II mattresses.

3/ Not applicable to mattresses using grade B felt (see 4.4.4.1.3 and 6.2).

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TABLE IV. Physical requirements - Type III and type IV

Size	Length inches ± 1/2 inch	Width inches ± 1/2 inch	Thickness inches ± 1/2 inch	Weight pounds (min)	No. of ventilators Each side	Total No. of coils (min)	Weight of felt layer on each face pounds 1/
1	76	29	6-1/2	30	4	126	8-1/2
2	78	33	6-1/2	30	4	157	10
3 2/		-	-	-	-	-	-
4	75	36	6-1/2	36	4	162	10-1/2
5	78	36	6-1/2	39	4	171	11
6	84	36	6-1/2	41	4	180	12
7	75	38	6-1/2	43	4	162	11-1/2
8	78	39	6-1/2	41	4	171	12
9	78	41-1/2	6-1/2	43	4	190	13
10	78	47	6-1/2	50	4	209	15
11	73	53	6-1/2	51	4	221	16
12	75	53	6-1/2	60	4	234	16
13	80	36	6-1/2	36	4	171	11
14	80	38	6-1/2	40	4	171	12-1/2
15	80	53	6-1/2	60	4	247	17-1/2
16	80	60	6-1/2	60	4	247	17-1/2

1/ Not a specification requirement, included for information purposes.

2/ Size 3 is not applicable to type III and IV mattresses.

3/ Not applicable to mattresses using grade B felt (see 4.4.4.1.3 and 6.2).

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3.7 Labels. All labels shall be sewn at one end of the mattress into the same edge seam.

3.7.1 Statutory label. Statutory labels shall conform to the legal requirements of the state in which the mattresses are manufactured.

3.7.2 Identification label. Each mattress shall have an identification label conforming to type I, class 5 of DDD-L-20, except that the requirement for launderability shall not apply. The size of characters of the inscription on the label shall be 3/32 inch to 1/4 inch. The spring unit supplier's name and type of spring unit along with the lot identification number (flammability) shall be included in the marking.

3.7.3 Medical label. When specified (see 6.2) mattresses procured for medical activities shall have a label conforming to type I, class 9 of DDD-L-20, bearing manufacturer's name or registered trade mark, FSN, item description and lot identification number (flammability).

3.8 Deviation of mattress surface (type I only). The maximum average deviation of mattress surface shall be not greater than 3 degrees and the maximum deviation of mattress surface at any place shall be not greater than 6 degrees, when tested as specified in 4.4.6.

3.9 Depression under compression-plunger (type I only). The difference between the minimum and maximum depression produced by the plunger across the mattress under loads of 20 pounds shall be not greater than 25 percent of the height of the mattress, when tested as specified in 4.4.6. The depression at any point shall be not greater than 60 percent of the height of the mattress.

3.10 Workmanship. The finished mattresses shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article sample submitted in accordance with 3.1 shall be inspected as specified in 4.4.3 for compliance with design, material, construction, workmanship, weight, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Testing of components. In addition to the quality assurance provisions of the referenced specifications, components and materials listed in table VI shall be tested for the characteristics shown therein. The methods of testing specified in FED-STD-191, wherever applicable, and as specified herein shall be followed. When the data in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table VI, they shall be as required in the referenced test method. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be in accordance with table V and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

TABLE V. Sampling for testing

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VI. Test methods of components

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as	Sample unit
Edge binding tape (1 spool)	Material identification	3.3.11.1	Standard commercial	1	Pass or fail	
	Breaking strength	3.3.11.1	6015 1/ 2/	-	-	1 cone, spool or 30 feet of tape
	Width	3.3.11.1	Rule	3	Pass or fail	
Tufting tape, braid or twine (1 cone or spool)	Material identification	3.3.11.2	Standard commercial	1	Pass or fail	
	Breaking strength	3.3.11.2	6015 1/ 2/	-	-	1 cone, spool or 30 feet of tape
	Width (when applicable)	3.3.11.2	Rule	3	Pass or fail	
Coils for type I mattress (100 coils)	corrosion resistance	3.3.18.1	4.5.3.2	3	Pass or fail	3 coils
Coils for types II, III and IV mattresses 100 coils or 1 complete spring unit assembly	Compression Set	3.3.18.2 or 3.3.18.3 or 3.3.18.4 (as applicable)	4.5.3.1 3/	3	Pass or fail	2 coils of each type or 1 complete spring unit assembly (as applicable)

1/ Refers to FED-STD-191.

2/ Except that the full width of the tape shall be used.

3/ A supplier's certificate of compliance shall be furnished and will be acceptable for the stated requirements.

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4.4.1.2 Components, materials and component assemblies. When the components, materials and component assemblies listed below are used in the fabrication of mattresses, a supplier's certificate of compliance (see 4.1.1) shall be furnished and will be acceptable for the specified requirements.

Component, material or component assembly	Requirement paragraph
Cotton 1/	3.3.1
Linters 1/	3.3.2
Polyester fiber	3.3.3
Felt 1/	3.3.4
Grade B felt (when applicable)	3.3.4.2
Barrier materials	3.3.5
Inner support tufting fabric	3.3.8
Heat shield or flange fabric	3.3.9
Handles	3.3.12.2
Tufting buttons	3.3.14
Ventilators	3.3.15
Wire mat components (wire and fiber cords)	3.3.16
Hog rings	3.3.17
Coils - Types I, II, III, and IV	3.3.18
Stabilizer wire (end rods)	3.3.20
Helicals	3.3.21
Weight of padding on border	3.5.9

1/ Unless otherwise specified (see 6.2), the cotton or linters or felt shall be examined, to determine conformance with 3.3.1, 3.3.2 and 3.3.4 inclusive, by classifiers licensed by the U. S. Department of Agriculture; or representative samples, each weighing not less than 1/2 pound, of the cotton or linters and a minimum 12 inches by 12 inches of felt shall be forwarded to the administrative office concerned at Washington, DC, where inspection by the proper board of the U. S. Department of Agriculture shall be obtained. The opinion of this board as to the true grade or quality will be final. Inspection reports from the U.S.D.A. shall be furnished and will be acceptable for the stated requirements.

4.4.2 Intermediate inspection. Intermediate inspection shall be performed on all innerspring units, except when units are listed on an ASL (Acceptable Suppliers List). Inspection shall also be performed on the wire mat when used as an insulator. The defects found during examination shall be in accordance with 4.4.2.1, 4.4.2.2, 4.4.2.3 and 4.4.2.4. The inspection levels and AQL's expressed in defects per 100 units shall be as follows:

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For examination in	Inspection levels	AQL's
4.4.2.1	S-3	one class 4.0
4.4.2.2	S-3	one class 4.0
4.4.2.3	S-3	one class 4.0
4.4.2.4	S-3	one class 4.0

4.4.2.1 Examination of innerspring assembly (type I mattress) for visual defects. The sample unit for this examination shall be one innerspring assembly protected by a fitted cover. Examination of innerspring assembly for defects in material and workmanship shall be as follows:

Examine	Defect
Sheeting or non-woven fabric	Any material defect such as a hole or smash. Any workmanship damage such as a cut, tear, rip, mend, burn or needle chew not repaired.
Construction and workmanship	Both upper and lower surfaces of assembly not covered by a cover. Cover does not overlap all edges of innerspring construction. Cover not attached to spring construction or to assembled border all around by twine stitching. Stitches do not engage and secure each border coil at sides and every other coil across ends of cover and assembled borders. Any stitch that is not knotted. More than two stitches not adequately knotted. Finish on spring unit not as specified. Finish missing from spring unit.
Coils	Any number of coils per row (in direction of length and width) less than specified. 1/

1/ The number of coils in each direction can be counted prior to fitting the cover to the individual pocketed coils.

4.4.2.2 Examination of the all-wire spring unit, free-end coil construction assembly (type II mattress) for visual defects. The sample unit for this examination shall be one type II mattress spring unit.

Examine	Defect
Construction	Transverse rows do not consist of alternating right-hand and left-hand coils. Alternating right-hand and left-hand coils are not connected with a helical running at right angles to the long axis of the unit in such a manner as to prevent lateral slippage.

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Examine	Defect
Construction (cont'd)	Free end of coil is not anchored within the helical. Intermediate convolutions of the individual coils in the longitudinal rows forming the side are not cylindrical in shape. Edges of all outside coils are not constructed so as to present a straight edge to the construction. The individual coils forming the end rows are not offset cylindrical or barrel type and are not overlapped to form closed end or bearing convolutions. Corners of stabilizer wires are die-formed to less than 3-1/2 inch or more than 4-1/2 inch radius. The free end of the stabilizer wire is not placed within each of the 4 end helicals and the free ends of the wire do not extend to a point or line with the center of the second row of coils at each end and not fastened so as to prevent slippage. Less than 6 convolutions on any free-end corner coil. Any free-end corner coil not cylindrical in shape. Any corner coil formed to less than 3-1/2 inch or more than 4-1/2 inch radius. Not an hourglass shape on any other coil not indicated above. Coil not indicated above having less than 4-1/2 active turns or complete convolutions (when counted from last point of contact of coil with helical on top to the first point of contact with the helical on the bottom). Coil having a height less than 4-3/4 inches or more than 5-1/4 inches. Total number of coils less than specified. End of helicals not turned in forming a smooth rounded termination. Finish on spring unit not as specified. Finish missing from spring unit.

4.4.2.3 Examination of the all-wire spring unit, knotted-coil construction (type III mattress), and offset construction (type IV mattress), for visual defects. The sample unit for this examination shall be one spring-unit as applicable for construction type for a type III mattress or a type IV mattress. Examination of the spring-unit for defects in material design and workmanship for the applicable construction type shall be as specified in tables VII and VIII.

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TABLE VII. All wire spring unit, knotted coil construction (type III mattress)

Examine	Defect
Construction	Spring unit does not consist of either transverse rows of coils connected by the use of helicals running at right angles to the long axis of the unit as specified; or the spring unit does not contain transverse rows of individual coils, or units of not more than 4 coils to a unit joined to the neighboring coils or units as specified. Coil contains less than three active turns or convolutions (including only portions between knots of the coil). Coil height is less than 4-3/4 inches or more than 5-1/4 inches. Corners of stabilizer wires are die-formed to less than 3-1/2 inches or more than 4-1/2 inches radius. Total number of coils less than specified. All knots not turned down. Any end of any helical not turned in so as to form a smooth and rounded termination of the helicals. Finish on spring unit not as specified. Finish missing from spring unit.

TABLE VIII. All wire spring unit, offset construction (type IV mattress)

Examine	Defect
Construction	Offset coils are not connected with a helical running at right angles to the long axis of the unit in such a manner as to prevent lateral slippage. Offset section of coil is not anchored within the helical. Corners of stabilizer wires are die-formed to less than 3-1/2 inches or more than 4-1/2 inches radius. The stabilizer wire is not fastened to prevent slippage and does not extend around the corners more than 10 inches. Less than 3 active turns or complete convolutions on any offset coil.

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TABLE VIII. All wire spring unit, offset construction (type IV mattress) (cont'd)

Examine	Defect
Construction (cont'd)	Coil having a height of less than 4-3/4 inches or more than 5-1/4 inches. Total number of coils less than specified. End of helical not turned and locked and forming a smooth rounded termination. Finish on spring unit not as specified. Finish missing from spring unit.

4.4.2.4 Dimensional examination of wire mat. The sample unit for this examination shall be one completely fabricated wire mat. Defects shall be as listed below:

Examine	Defect
Dimensions (length and width)	Not as specified
Spacing of steel wires	Not as specified
Spacing of fiber cords	Not as specified

4.4.3 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.

Operation or assembly	Requirement paragraph
Application of insulation on innerspring units	3.5.5
Application of padding (upholstery) to innerspring units	3.5.6
Application of barrier material (toppers) to spring units (when applicable)	3.5.7
Construction of cover (with or without heat shield or flange)	3.5.8
Construction of border	3.5.9
Construction of edges	3.5.10
Concealed operations of innertufted mattresses	3.5.11, 3.5.11.1 and 3.5.12

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4.4.4 Inspection of the end item.

4.4.4.1 Visual examination of mattresses. The defects found during examination shall be classified in accordance with 4.4.4.1.1, 4.4.4.1.2, and 4.4.4.1.3. The lots shall be inspected in accordance with 4.4.4.1.4. The sample unit for these examinations shall be one mattress (as applicable). Any mattress removed from the polyethylene bag for examination need not be resealed. The open end of the polyethylene bag shall be repaired (taped up) prior to placing the packaged mattress (mattress in bag) back in to a carton.

4.4.4.1.1 Examination of mattress for defects in design, material, construction and workmanship.

Examine	Defect	Classification Major	Minor
Quality of ticking	Hole, cut, tear, smash, mend or open place:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Loose yarn, snagged thread, misweave, missing yarn, broken yarn, thin place, shade bar, or woven in waste:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Rancid or other objectionable odor	X	
	Not fabric specified	X	
	Start marks (heavy or light)		X
	Spot or stain 1/4 inch or over in diameter, clearly noticeable when viewed at a distance within 3 feet		X
	Excessive neppiness	X	
Seams and stitching	Any required seam or stitching operation omitted	X	
	Ticking not seamed with seam type SSa-1 or LSb-2		X
	Not stitch type 301 or 401 (as applicable)		X
	Stitches per inch less than specified	X	
	Stitches per inch more than specified		X
	Ends of any seam not backstitched (except when end of seam is caught in another seam or stitching, or when closing the mattress) as specified		X
	Raw edge more than 1 inch in length		X
	Open seam more than 1 inch in length	X	

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Examine	Defect	Classification Major	Minor
Seams and stitching (cont'd)	Open seam 1 inch or less in length		X
NOTE:	A seam shall be classified as open when one or more stitches are broken or when two or more continuous skipped stitches or run-offs occur.		
	Stitch tension loose, resulting in a loosely secured seam		X
	Stitch tension tight (stitches break when normal strain is applied to the seam or stitching)		X
NOTE:	Puckering is evidence of tight tension, when puckering is evident, the inspector shall test seams by exerting pull on seam.		
	Needle chew, hole or cut:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Seam badly puckered, pleated, twisted or caught in unrelated operation		X
	Five or more untrimmed thread ends or loose threads, 1 inch or more in length		X
Construction	Not type specified	X	
	Sagging appearance, i.e., filling not evenly distributed	X	
	Lumps clearly noticeable	X	
	Thin hollow places clearly noticeable	X	
	Border or corner contains insufficient padding	X	
	Spring coils can be felt through ticking	X	
	Face-panel (top or bottom) ticking more than two pieces of material	X	
	Blue stripe of ticking does not run in lengthwise direction		X
	Ticking stripes not properly matched adversely affecting appearance		X
	Corner(s) not firm	X	
Border and bound edges	Border not constructed as specified	X	
	Border fabricated with more than two pieces of ticking (all around)		X

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Examine	Defect	Classification Major	Minor
Border and bound edges (cont'd)	Transverse stitching on border does not extend across the full width of the border		X
	Transverse stitching on border less than 1-1/4 inch on centers or more than 1-3/4 inch on centers		X
	Not box edge or box edge not sharply defined		X
	Binding tape does not extend completely around edge of mattress	X	
	More than 3 pieces of binding tape on edge of mattress	X	
	Binding tape puckered, pleated, or twisted		X
	Color of tape not as specified		X
	Ventilators insecurely attached or not firmly anchored to border		X
	Less than four ventilators in each side	X	
	Evidence that ventilator passage is obstructed to an extent interfering with free passage of air into the spring section		X
	Any ventilator not aligned halfway between the top and bottom of border (permit ± 1/2 inch tolerance).		X
	Ventilators not equidistant on either side of ends of mattresses (permit ± 1 inch tolerance)		X
	Ventilator screen damaged		X
	Less than two handles on each side of mattress (when applicable)	X	
	Any handle misplaced, or misaligned		X
	Horizontal handle pulls out of retainer (check by turning mattress over, using all handles)	X	
	Horizontal handle grommet or eyelet insecurely attached to border	X	
Tufting	Not type specified	X	
	Not accomplished as specified		X
	Tufts do not appear on both sides of mattress	X	
	Less than the number of tufts specified in table I		X
	Knots of tape tying tufts (when applicable) appear above surface on either side of mattress		X

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Examine	Defect	Classification Major	Classification Minor
Tufting (cont'd)	Tear over 3/4 inch in ticking under button (check underneath a minimum of three buttons on each side of mattress)		X
	Arrangement of diamond pattern not as specified		X
	Spacing button tufts not uniform affecting appearance		X
	Depth of tufts not as specified		X
	NOTE: Check a minimum of three tufts by inserting a pointed thin piece of metal or wire, and of sufficient length, through the tuft until the point is felt on the bottom side of mattress. Grasp the instrument at the beginning of the depth of the top tuft and remove. Measure the distance from point of instrument to where thumb and forefinger are holding the instrument.		
Quilting	Any thin place devoid of padding 2 inches square or more in area	X	
	Any thin place devoid of padding, more than 1 inch square in area, but less than 2 inches square in area		X
	Three or more thin places devoid of padding regardless of area	X	
	Any furrow, lump, bulge resulting from uneven or improper distribution of the padding	X	
	Five or more untrimmed thread ends or loose threads, one inch or more in length	X	
	Surface twisted, pleated, wavy or puckered	X	
	A runback of more than 1-1/2 inches of broken or missing stitches in the quilting		X
	Any splice in ticking cloth	X	
	Needle chews:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Any area evidencing complete bobbin change		X
	More than 1-1/2 inches of broken or missing stitches in quilting	X	
	One or more rows of stitching omitted in quilting	X	
	Stitches per inch:		
	One or more stitches less than specified	X	
	One or more stitches more than specified		X

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Examine	Defect	Classification Major	Minor
Labels	Missing, inclomplete, incorrect, or illegible	X	

4.4.4.1.2 Examination of mattress for defects in dimestions.

Examine	Defect
Thickness	Average is less than 6 inches Average is more than 7 inches
Length	Average length is beyond tolerances indicated in tables II, III and IV
Width	Average width is beyond tolerances indicated in tables II, III and IV
Radius at corners of mattress	Less than inches More than 5 inches
Distance of outer tufted row to end of mattress	Less than 5 inches More than 8 inches
Distance of outer tufted row to edge of mattress	Less than 5 inches More than 7 inches

4.4.4.1.3 Examination of mattress for defects in weight.

Examine	Defect
Weight (as applicable) 1/	Less than minimum weight specified in tables II, III and IV

1/ Weight. Weight may be determined under prevailing atmospheric conditions, except in the settlement of disputes. In case of dispute, weight shall then be determined under standard atmospheric conditions as defined in FED-STD-191. Deviations from the minimum specified weight requirements due to the use of grade B felt is subject to the approval of the contracting officer.

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4.4.4.1.4 Inspection levels and acceptable quality levels (AQL's). Inspection levels and AQL's expressed in defects per hundred units shall be as follows:

For examination in	Inspection level	AQL
4.4.4.1.1	II	Major 4.0 Total 10.0
4.4.4.1.2	S-3	1 class 4.0
4.4.4.1.3	S-3	1 class 4.0

4.4.5 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-M-3996 to determine that packaging, packing and marking comply with section 5 requirements of this specification.

4.4.6 Testing of the end item. Testing of the completely fabricated mattress shall be performed in accordance with the provisions of table IX for the characteristics shown therein. The sample unit shall be one completely fabricated mattress. The sample size (number of sample units) shall be in accordance with MIL-STD-105. All test reports shall contain the individual values utilized in expressing the final result.

TABLE IX. Test methods for mattresses

Characteristics	Requirement para.	Test method	Number of determinations per sample unit	Results reported as	Insp. level	AQL
Folding of mattress freehinge action (type III mattress)	3.3.19.3	4.5.4	1	Pass or fail	S-3	4.0
Deviation of mattress surface (type I mattress)	3.8	4.5.1	10	To nearest degree	S-3	4.0
Depression under compression plunger (type I mattress)	3.9	4.5.2	15 (5 in each of 3 parallel lines	To nearest 1.0 percent	S-3	4.0

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4.5 Test methods.

4.5.1 Tripod-level test (type I mattress). The uniformity of mattress compression shall be determined by the use of a swivel level mounted on a tripod base. The instrument shall be placed on the mattress and oriented at random at 10 or more places to indicate, in degrees, the maximum deviation of the mattress top from a horizontal plane. During the test no leg of the tripod shall be located directly on a tuft nor less than 1-1/2 inches from the edge of the mattress. The tripod level consists of a base plate supported by three 1-1/4 inch ball-pointed legs 3 inches long and 6 inches center to center. Upon the base plate is mounted a 9-inch quadrant divided into forty-five 1 degree divisions, each side of a plumbline center. A plumb bob is suspended from the apex of the quadrant. The quadrant is swivel mounted in the center of the base plate. The weight of the tripod level complete is 10 pounds.

4.5.2 Compression-plunger test across mattress (type I mattress). This test consists in compressing the mattress with a ball-pointed plunger 1-1/4 inches in diameter using a 20 pound load on the plunger. The depth of depression is noted at intervals of 1-1/4 inches in three parallel lines 1-1/4 inches apart across the mattress in right-angle directions.

4.5.3 Coils.

4.5.3.1 Coil compression. The height of the coil shall be measured prior to determining coil compression. Coil compression shall be accomplished by placing either end of a single coil on a flat surface and then compressing it manually, using a board or block over the top of the coil, until all adjacent convolutions are touching or are compressed to a height of not more than 2 inches. This procedure shall be repeated 20 times to determine conformity with the requirements specified in 3.3.18.2, 3.3.18.3 and 3.3.18.4. If a large number of coils must be tested, a simple mechanical reciprocating type testing device may be used to accomplish the above test.

4.5.3.2 Salt-spray test. The sample coils for type I mattresses shall be subjected to a salt-spray test consisting of exposure in a closed tank of neutral material, such as wood or soapstone, to a finely divided spray at room temperature (65° to 80°F.) of 4 or 5 percent sodium chloride (by weight) in distilled water for a period of 72 hours. The tank shall be so designed so that there will be no direct impingement of the spray upon the test specimens, and the spray shall circulate freely about all specimens to the same degree. None of the test specimens shall show signs of failure of protective coating, that is, the appearance of rust on the steel.

4.5.4 Folding (type III). The mattress shall be folded together, end to end and then opened flat again. The mattress shall be observed for lateral slippage of coils, an inability to permit a free hinge action or any visible physical distortion.

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4.6 Flammability. Flammability tests shall be performed in accordance with all the provisions of the Consumer Product Safety Commission Flammability Standard for Mattresses FF-4-72. A supplier's certificate of compliance shall be furnished and will be accepted for the stated requirements. Certification shall include construction and material details used to meet the standard.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Mattresses shall be packaged in accordance with the applicable requirements of MIL-M-3996.

5.1.2 For civil agency procurements, where level B packaging is specified in the contract or order, the requirements in 5.1.1 (level A) shall apply.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Mattresses shall be packed in accordance with the applicable requirements of MIL-M-3996.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123.

5.3.1.1 Special marking. The following marking shall also appear on at least two side edges of each shipping container in letters not less than 1-1/2 inches in height.

STOW FLAT-DO NOT STOW ON END OR SIDE

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-M-3996.

6. NOTES

6.1 Intended use. These mattresses are intended for use by civil and military agencies of the Government.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and size of mattress required (see 1.2).
(c) First article. When preproduction sample is required (see 3.1, 4.1.1 and 6.6).
(d) When grade B felt is required (see 3.3.4.2).

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(e) Whether medical label is required (see 3.7.3).
(f) When examination of cotton linters or grade A felt is not required to be examined by the U.S. Department of Agriculture.
(g) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one finished mattress. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Flame retardant and smolder resistant finishes and materials. Approval of such components and combinations is the responsibility of the U.S. Army Natick Laboratories, Natick, MA 01760, and is based on extensive tests including those for toxicity performed by Office of the Surgeon General which are not set forth in this specification. Because of the time necessary to conduct full evaluation, only those chemical treatments and materials already approved and so listed in the invitation for bid or request for proposal shall be considered acceptable for the related procurement.

6.5 Bonding agents. Approval of bonding agents for use in battings covered by this specification is the responsibility of the U.S. Army Natick Laboratories, Natick, MA 01760. Approval is based on more extensive evaluations, including those of toxicity, which are not set forth in the specification. Because of the time necessary to conduct full evaluation, only those bonding agents already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

6.5.1 Submittal for approval. Request for approval should be submitted to the U.S. Army Natick Laboratories, attention of STSNL-TDO. It should be accompanied by approximately five square yards of a minimum of 2 oz/sq yd polyester batting manufactured with the proposed bonding agent together with applicable test data. Information is to be furnished regarding the commercial designation and chemical description of the bonding agent. (Approved lists will indicate only the commercial designation). Chemical identification with products already approved should be made, if applicable, and any available data on skin irritancy or toxicity should be submitted. If applicable, a statement should be included that as used the resin is well cured and that no materials with known high toxicity are used prior to cure.

6.6 Supersession data. This specification includes the requirements of Interim Purchase Description IP/DES-3-3 dated 19 April 1973.

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Custodians:

Army - GL
Navy - YD
Air Force - 82

Review activities:

Army - MD
Navy - SA, MC
Air Force - 45
DSA - DM

User activity:

Navy - SH

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS
JUS-FPI
HEW-FDA
INT-BPA
COM-NBS
AGR-AMS

Project No. 7210-0173

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

Mattresses, Bed, Innerspring V-M-96G

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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